Lab Exercise 1- Docker Fundamental Commands

Name-Manish kumar

SAP ID- 500119723

Batch-2(DevOps)

Objective

Learn and practice the fundamental Docker CLI commands to:

- Manage images and containers
- Understand container lifecycle
- Build and run applications using Docker

---

1. Setup

Prerequisites

- Docker installed on your system

  (Check using: docker --version)

- Internet access to pull images

---

2. Basic Docker Commands

Step 1: Verify Installation

```
docker --version

docker info
```

```
C:\Users\Misha>docker --version
Docker version 28.5.1, build e180ab8

C:\Users\Misha>docker info
Client:
 Version:    28.5.1
 Context:    desktop-linux
 Debug Mode: false
 Plugins:
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.9.11
    Path:     C:\Program Files\Docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.29.1-desktop.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v2.40.2-desktop.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.45
    Path:     C:\Program Files\Docker\cli-plugins\docker-debug.exe
  desktop: Docker Desktop commands (Docker Inc.)
```

Expected Output: Docker version and system details.

Step 2: Pull an Image from Docker Hub

```
docker pull ubuntu:latest
```

```
C:\Users\Misha>docker pull ubuntu:latest
latest: Pulling from library/ubuntu
a3629ac5b9f4: Pull complete
Digest: sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
```

Explanation: Downloads the latest Ubuntu image from Docker Hub.

Check available images:

```
docker images
```

```
C:\Users\Misha>docker images
REPOSITORY                        TAG       IMAGE ID       CREATED        SIZE
ubuntu                            latest    cd1dba651b30   5 days ago     117MB
smartcampus-frontend              latest    584edc3d6c7f   7 weeks ago    85MB
misha1207/smartcampus-frontend    latest    584edc3d6c7f   7 weeks ago    85MB
<none>                            <none>    177318997aef   7 weeks ago    84.9MB
smartcampus                       latest    bc572bac8bbe   7 weeks ago    199MB
misha1207/smartcampus             latest    bc572bac8bbe   7 weeks ago    199MB
nginx-html-app                    latest    f2185d94d89c   2 months ago   225MB
sonarqube                         lts       f709975ab31d   4 months ago   1.02GB
hello-world                       latest    f7931603f70e   5 months ago   20.3kB
```

Step 3: Run a Container

Run Ubuntu interactively:

```
docker run -it --name myubuntu ubuntu bash
```

```
C:\Users\Misha>docker run -it --name myubuntu ubuntu bash
root@93542becf8ad:/#
```

Now you're inside a running container (prompt will change).

Exit:

```
Exit
```

```
C:\Users\Misha>docker run -it --name myubuntu ubuntu bash
root@93542becf8ad:/# exit
exit

C:\Users\Misha>
```

Step 4: List Containers

- Show running containers:

```
docker ps
```

```
C:\Users\Misha>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\Misha>
```

- Show all containers (including stopped):

docker ps -a

```
C:\Users\Misha>docker ps -a
CONTAINER ID   IMAGE           COMMAND                  CREATED              STATUS                          PORTS
                NAMES
93542becf8ad   ubuntu          "bash"                   About a minute ago   Exited (0) About a minute ago
                myubuntu
d780008ef8b3   177318997aef    "/docker-entrypoint.…"   7 weeks ago          Exited (0) 7 weeks ago
                youthful_gagarin
466a1f3515b4   177318997aef    "/docker-entrypoint.…"   7 weeks ago          Created
                great_williams
3a844f4e1cb3   smartcampus     "docker-entrypoint.s…"   7 weeks ago          Exited (255) 7 weeks ago        0.0.0.0:300
0->3000/tcp    youthful_lichterman
6ec9ca2fd92a   smartcampus     "docker-entrypoint.s…"   7 weeks ago          Created
                quizzical_davinci
c4d7dc365d02   hello-world     "/hello"                 2 months ago         Exited (0) 2 months ago
                gifted_gates
3be0342fd4e2   sonarqube:lts   "/opt/sonarqube/dock…"   2 months ago         Exited (255) 8 weeks ago        0.0.0.0:900
0->9000/tcp    sonar
```

Step 5: Start / Stop / Remove Containers

Start a stopped container

docker start <container_id>

```
C:\Users\Misha>docker start 93542becf8ad
93542becf8ad
```

Stop a running container

docker stop <container_id>

```
C:\Users\Misha>docker stop 93542becf8ad
93542becf8ad
```

Remove a container

docker rm <container_id>

```
C:\Users\Misha>docker rm c4d7dc365d02
```

Tip: Use docker ps -a to get container IDs.

Step 6: Remove Images

docker rmi ubuntu:latest

```
C:\Users\Misha>docker rmi ubuntu:latest
Error response from daemon: conflict: unable to delete ubuntu:latest (must be forced) - container 93542becf8ad is using
its referenced image cd1dba651b30
```

Note: You must stop and remove all containers using that image first.